

Department Overview

- **Department Name:** Department of Computer Engineering
- **Faculty:** Faculty of Information & Communication Technology (FICT)
- **Accreditation:** The program is fully accredited by the **Pakistan Engineering Council (PEC)**.
- **Core Focus:** Blends hardware engineering and computer science concepts. It covers software programming, hardware design, industrial automation, embedded systems, robotics, multimedia, and artificial intelligence.

Vision and Mission

- **Vision:** To produce professional Computer Engineers who can use knowledge, technical expertise, and entrepreneurial abilities to compete with upcoming challenges in computing fields while adhering to highest standards of morality and ethics.
- **Mission:** To prepare graduates to accept and fulfill responsibilities across a broad spectrum of activities, including analysis, synthesis and design of the modern computing systems, serve government and industry in various roles and contribute in academia and research.

Bachelor of Science (BS) Program

Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering) / ICS with Mathematics & Physics, or an equivalent qualification with at least **60% marks**. Alternatively, a Diploma of Associate Engineering (DAE) in a relevant field with at least **60% marks**.
- **Entry Test:** Candidates must clear an entrance test conducted by NTS.

Degree Requirements

- **Total Credit Hours:** 134
- **Total Courses:** 43
- **Total Labs:** 21 or 26
- **Minimum Academic Standing:** CGPA $\geq$ 2.0

Curriculum Plan (Program Schema by Semester)

Semester	Course Code	Course Title	Prerequisite	Credit Hours
1st	CS 118 / 118L	Computer Fundamentals	None	2 + 1

		& Lab		
	PHY 205 / 205L	Applied Physics & Lab	None	2 + 1
	MATHP 118	Calculus & Analytical Geometry	None	3
	HUM 101	Islamic Studies/Ethics	None	2
	ENG 161	Reading and Writing Skills (Eng.-I)	None	3
	ENGG 202L	Workshop Practice Lab	None	1
	ENVIRON-102	Occupational Health and Safety	None	1
2nd	CS 112 / 112L	Computer Programming & Lab	None	3 + 1
	MATHP 111	Linear Algebra	None	2
	EE 221 / 221L	Electronic Circuits and Devices & Lab	None	3 + 1
	ENG 361	Communicatio	None	2

		n Skills		
	EE 203 / 203L	Linear Circuit Analysis & Lab	None	3 + 1
	HUM 102	Pakistan Studies and Global Perspective	None	2
3rd	TE 305 / 305L	Computer Comm. and Networks & Lab	None	3 + 1
	CS 212 / 212L	Object-Oriented Programming & Lab	Computer Programming + Lab	3 + 1
	MATHA 213	Complex Variables and Transforms	None	3
	EE 110 / 110L	Digital Logic Design & Lab	None	3 + 1
	MATHA 234	Discrete Structures	None	3
4th	EE 261 / 261L	Signals & Systems & Lab	Complex Variables and Transforms	3 + 1

	CE 211 / 211L	Computer Architecture and Organization & Lab	Digital Logic Design + Lab	3 + 1
	MATHA 214	Differential Equations	Calculus & Analytical Geometry	3
	CS 214 / 214L	Data Structures & Algorithms & Lab	Object-Oriented Programming + Lab	3 + 1
	—	MS-Elective I	None	3
5th	CE 326 / 326L	Microprocessors and Interfacing & Lab	Computer Architecture and Org. + Lab	3 + 1
	CS 341 / 341L	Operating Systems & Lab	Object-Oriented Programming + Lab	3 + 1
	TBA	Social Science-Elective I	None	2
	HUM 292	Technical Writing	None	3
	TE 412 / 412L	Digital Signal Processing &	Signals &	3 + 1

		Lab	Systems + Lab	
6th	TBA	CEDE - I & CEDE - II	TBA	Var (2+1 or 3)
	MATHA 217	Probability Methods in Engineering	None	3
	CS 331 / 331L	Database Management Systems & Lab	None	3 + 1
	EE 302 / 302L	Digital System Design & Lab	Digital Logic Design + Lab	3 + 1
7th	TBA	MDEE – I & CEDE-III	TBA / None	Var
	TBA / TBA	Numerical Analysis & Lab	TBA	2 + 1
	CE 451L	Senior Design Project-I	None	3
	CE 303 / 303L	Software Engineering & Lab	None	2 + 1
8th	TBA	CEDE - IV	TBA	Var
	TBA	Management Science Elective-II	None	3

	TBA	MDEE – II	None	3
	CE 452L	Senior Design Project-II	Senior Design Project-I	3
	TBA	Social Science Elective-II	None	3

Master of Science (MS) Program

Admission Requirements

- **Academic Background:** 16 years of education or equivalent (e.g., B.E or BS 4-year degree) in a relevant field from an HEC-recognized university. Applicants must have at least **60% marks** under the annual grading system or a minimum CGPA of **2.5 out of 4.0**.
- **Entry Test:** Must clear the admission test conducted by the University or its authorized testing agency with a minimum cumulative score of **50%**.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Courses:** 10
- **Minimum Academic Standing:** CGPA $\geq$ 3.0